

Hypothesis Testing

> Null Hypothesis (H_0):

There is no difference between two means. It shows equality.

> Alternative Hypothesis (H_1):

1) One Tailed Test

2) Two Tailed Test



Q: The school principal wants to test if it is true what teachers says that high school juniors use the computer an average 3.2 hours a day. what are null and alternate hypothesis.

Ans: Null hypothesis:- There is no difference.

$$H_0: \mu = 3.2$$

Alternate Hypothesis:- $H_1: \mu \neq 3.2$ (Two-tail)

Q: A deodorant brand claims fragrance stays more than 2.2 hours.

Ans: Null Hypothesis:- $H_0: \mu \geq 2.2$

Alternate Hypothesis:- $H_1: \mu < 2.2$ (one-tail)

$$H_0: \mu \geq 2.4$$

$$H_1: \mu < 2.4 \text{ hrs}$$



Q: A student believes she scores less than the avg student who gets 75 marks. Null & Alternative Hypothesis?

$$H_0: \mu \geq 75$$

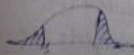
$H_1: \mu < 75$ (one-Tail)



Q: You want to test if boys and girls have different avg study times. Null & Alternative Hypothesis?

$$H_0: \mu_b = \mu_g$$

$$H_1: \mu_b \neq \mu_g$$



Q: A company pass the quality

$$H_0: \mu \geq$$

$$H_1: \mu <$$

1) Confidence Interval

2) Significant

Procedure

1) Define or

2. Select sign

3. choose th

4. calculate

5. Compare

P-value

6. Draw t

If p

If p

rej

Confidence

C.I

Confidence

C.I

Q:- A company claims that atleast 90% of their products pass the quality test

$$H_0: \mu \geq 90$$

$$H_1: \mu < 90$$

1) Confidence Interval

2) Significant Test:-

Procedure/test for significant Test:-

1) Define or frame the null and alternative hypothesis

2. Select significance level (α value) given in question

3. choose the appropriate test (z-test, t-test, paired t / chi, square test)

4. Calculate the test statistic value.

5. Compare the statistic value with critical value or p-value.

6. Draw the conclusion.

If p value $\leq \alpha$, reject null hypothesis

If p value $> \alpha$, ~~fromer to re~~ fail to reject / you cannot reject null hypothesis.

Confidence Interval for population mean.

$$C.I = \bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$$

Confidence Interval for population proportion

$$C.I = \bar{p} \pm z_{\alpha/2} \sqrt{\frac{p \cdot q}{n}}$$

$$p =$$

$$q = 1 - \bar{p}$$

[illegible]

(Faint handwritten notes visible through the paper)

(Faint handwritten notes on lined paper)

(Faint handwritten notes visible through the paper)

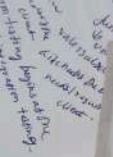
[illegible]

(Faint handwritten notes visible through the paper)

(Faint handwritten notes visible through the paper)

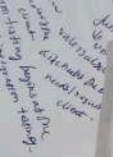
(Faint handwritten notes)

[illegible]

[illegible]

(Faint handwritten notes)

(Faint handwritten notes visible through the paper)



1. fasting begins with the
 sunset
 2. fasting begins with the
 sunset
 3. fasting begins with the
 sunset

[illegible]

(Faint handwritten notes visible through the paper)

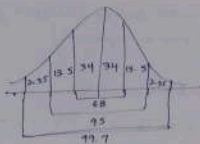
(Faint handwritten notes visible through the paper)



the proportion of
samples
customers.
ash, calculate
of customers

$$\frac{0.4}{12}$$

ad.



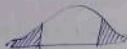
1. One-Tailed:

Rejection region is one side



2. Two-Tailed:

Rejection region on both sides



$$\frac{\sqrt{(0.7)(0.3)}}{100}$$

is true
by

Type-I Error: Conditional probability of rejecting a null hypothesis when it is true is called Type-I error.

> It is defined denoted by ' α ' (significance value).

> Also called as false positive.

Type-II Error: Conditional probability of failing to reject a null hypothesis when the alternative hypothesis is true is called Type-II error (or) False negative.

> It is denoted by ' β '.

1. Average annual salary of ml experts is different for males & female. Frame null & alternative hypothesis

$$H_0: \mu_m = \mu_f$$

$$H_1: \mu_m \neq \mu_f \rightarrow \text{2-Tailed since}$$



2. On an average people with PhD in analytics earn more than people with PhD in engineering.

$$H_0: \mu_A \leq \mu_E$$

$$H_1: \mu_A > \mu_E \quad \text{right-tailed}$$



3. Average life of vegetarians is more than meat eaters

$$H_0: \mu_V \leq \mu_M$$

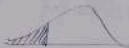
$$H_1: \mu_V > \mu_M$$



4. The salary of ML experts on average is atleast 100000\$ frame H_0 & alternative hypothesis.

$$H_0: \mu_{ML} \geq 100000\$$$

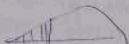
$$H_1: \mu_{ML} < 100000\$$$



5. Average waiting time at the London Heathrow Airport security check is less than 30 min.

$$H_0: \mu_{sc} \geq 30 \text{ min}$$

$$H_1: \mu_{sc} < 30 \text{ min}$$



6. Average annual salaries of male & female MBA students are different at the time of graduation.

$$H_0: \mu_M = \mu_F$$

$$H_1: \mu_M \neq \mu_F$$

Hypothesis Test

1) Z-Test:

$n \geq 30$, when u

i) one sample

ii) two sample

2) T-Test:

unknown σ

i) one-sample

ii) two-sample

iii) two-sample

iv) paired t-

3. chi-squared

> based on c

4. F-Test

> based on v

> Anova test

One Sample

1) The popula

2) The popu

3) $n \geq 30$

Q: An agen

monthly d

Bangalore i

from a re

disposable

the popul

test at

Hypothesis Test

1) Z-Test:

$n \geq 30$, when we know population standard deviation.

i) One sample Z-Test

ii) Two sample Z-Test

2) T-Test:

unknown σ & $n < 30$

i) One-sample T-Test

ii) Two-sample T-Test with equal variance

iii) Two-sample T-Test with unequal variance.

iv) Paired t-Test

3. chi-squared.

> based on categorical data

4. F-Test

> based on variance

> Anova test

One Sample z-test assumptions

1) The population standard deviation is known

2) The population follows the normal probability distribution

3) $n \geq 30$.

Q: An agency based out of Bangalore claimed that the average monthly disposable income of families living in Bangalore is greater than INR 4200 with a sd of INR 3200 from a random sample of 40000 families. The average disposable income was estimated as INR 4250. Assume that the population sd is INR 3200. Conduct an appropriate hypothesis test at 95% confidence level, to check the validity of the claim by agency.

Sol: $n=40000$, $M=4200$, $\bar{X}=4250$, $\sigma=3200$, $CI=95\%$, $z_{0.05}=1.96$

$$H_0: \mu \leq INR\ 4200$$

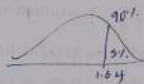
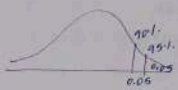
$$H_1: \mu > INR\ 4200$$

ii) Decide the α -value.

$$95\% \rightarrow \alpha \text{ value} = 0.05$$

$$Z_{\text{statistic}} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{4250 - 4200}{\frac{3200}{\sqrt{40000}}} = \frac{50}{8} = 3.125$$

iii) Compare statistic value with critical value.



$$3.125 > 1.64 \text{ (or)}$$

probability of Z Statistic $\leq \alpha$ comparison

$$1 - 0.9999 > 0.05$$

$$0.0008 < 0.05$$

reject the Null Hypothesis.

Q:- A passport office claims that the passport applications are processed within 30 days of submitting the application form. All necessary documents are:

The processing time of 40 passport applicants. The population standard deviation of the processing time is 12.5 days.

Conduct a hypothesis test at significance level $\alpha = 0.05$ to verify the claim made by the passport office.

→ passport processing time
16, 18, 30, 37, 28, 22, 19, 24, 35, 28, 29, 18, 23, 16, 24, 38, 26, 24, 11 = 40, $\bar{x} = 12.5$ days.

$$\bar{X} = \frac{1082}{40} = 27.05$$

$$H_0: \mu \leq 30$$

$$H_1: \mu > 30$$

2) Decide α value

$$95\% \rightarrow \alpha = 0.05$$

$$3) Z_{\text{statistic}} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

4) Compare Z_{stat}

$$-1.49 < 1.645$$

$$\uparrow$$

$$0.93189 > 0.05$$

5) Failed to reject

→ If population standard deviation is known

Q:- A manufacturer

of exactly 1 inch bolts and

should the

1 inch be

→ passport processing time

16, 16, 30, 37, 35, 22, 19, 35, 27, 32, 34, 28, 24, 35, 34, 21, 32,
29, 24, 35, 28, 29, 18, 31, 26, 33, 32, 24, 25, 22, 21, 27, 41, 23,
23, 16, 24, 38, 26, 28.

$n = 40$, $\bar{x} = 12.5$ days, $\alpha = 0.05$, $\mu = 30$ days.

$$\bar{x} = \frac{1082}{40} = 27.05 \text{ days}$$

1) $H_0: \mu \leq 30$

$H_1: \mu > 30$ (right tailed)

2) Decide α value

95% $\Rightarrow \alpha = 0.05$

3) $Z_{\text{statistic}} = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{27.05 - 30}{\frac{12.5}{\sqrt{40}}} = -1.49$

4) Compare $Z_{\text{statistic}}$ & α value

$-1.49 < 1.645$

↑

$0.93189 > 0.05$

probability of $-1.49 = 0.06811$

$1 - 0.06811$

$= 0.93189$

5) Failed to reject null hypothesis.

→ If population standard deviation is not given then consider sample standard deviation if sample $n \geq 30$.

Q:- A manufacturer produces bolt with a different thickness of exactly 1 inch. A customer takes a random sample of 100 bolts and find that $\bar{x} = 1.2$ inches & $s = 0.40$ inches. Should the manufacturer claim that bolts are exactly 1 inch be rejected. tested $\alpha = 0.01$.

sol: $n=100, s=0.40, \alpha=0.01$

$\bar{x} = 1.2, \mu = 1 \text{ inch}$

1) $H_0: \mu = 1$

$H_1: \mu \neq 1$ (two tailed test).

2) Decide α value 99%. $\alpha = 0.01$

3) $Z_{\text{statistic}} = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{1.2 - 1}{\frac{0.40}{\sqrt{100}}} = \frac{0.2}{\frac{0.40}{10}} = \frac{2}{0.40} = \frac{200}{40} = 5$

4) Compare $Z_{\text{statistic}}$ & α value

$5 > 2.58 \Rightarrow$ is falling in rejection region.

if Z -statistic is greater than 5, then consider its probability as 0. $0 < 0.01$.

we have enough

5) we have enough evidence to reject null hypothesis.

C.I = $\bar{x} \pm Z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$

$= 1.2 \pm (2.576) \cdot \frac{0.40}{\sqrt{100}}$

$= 1.2 \pm 2.576 \times \frac{0.40}{10}$

$= 1.2 \pm \frac{1.0304}{10} = 1.2 \pm 0.10304$

$1.09696 < \mu < 1.30304$

Q: A company claims that a box of it resin brand cereal contains at least 100 resins. An inspector working for the federal trade commission takes a random sample of 324 boxes of cereal & finds that $\bar{x} = 97.5$. 9 resins should the company claim to be rejected, tested $\alpha = 0.05$.

sd- Given that,

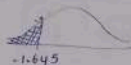
$$n = 324, \bar{x} = 97.5, s = 9, \alpha = 0.05, \mu = 100$$

1) Hypothesis:

$$H_0: \mu \geq 100$$

$$H_1: \mu < 100$$

2) Divide α value $\Rightarrow 0.05/2, \alpha = 0.05$



$$3) Z\text{-statistic} = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{97.5 - 100}{\frac{9}{\sqrt{324}}} = \frac{-2.5}{\frac{9}{18}} = \frac{-2.5}{0.5} = -5$$

$$P(5) = 0$$

4) Compare Z-statistic with α -value.

$$-5 < -1.645$$

$$0 < 0.05$$

5) Reject null hypothesis.

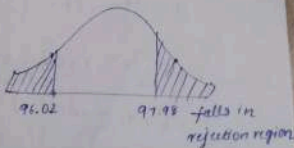
$$C.I = \bar{x} \pm Z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

$$= 97.5 \pm (1.96) \cdot \frac{9}{\sqrt{324}}$$

$$= 97.5 \pm 1.96 \times \frac{9}{18}$$

$$= 97.5 \pm 0.98$$

$$96.02 < \mu < 97.98$$



Q:- A researcher claims that 10 years old watch 2.6 hours of TV daily. you try to verify this with the following sample data
 $n=100$, $\bar{x}=6.1$ hrs, $s=2.5$ hrs, Test at $\alpha=0.01$

Sol:- Given that.

$$n=100, \bar{x}=6.1 \text{ hrs}, \mu=6.6 \text{ hrs}, s=2.5, \alpha=0.01$$

1) Hypothesis:

$$H_0: \mu = 6.6 \text{ hrs}$$

$$H_1: \mu \neq 6.6 \text{ hrs}$$

2) Decide α value $\Rightarrow 99\%$, $\alpha=0.01$



$$3) Z\text{-statistic} = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$= \frac{6.1 - 6.6}{\frac{2.5}{\sqrt{100}}} = \frac{-0.5}{\frac{2.5}{10}} = \frac{-5}{2.5} = -2$$

4) Compare Z statistic with α -value

$$-2 < -2.58$$

$$0.02275 < \frac{0.01}{2} = 0.005$$

5) failed to reject null hypothesis

$$C.I = \bar{x} \pm Z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

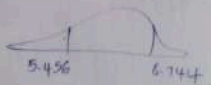
$$= 6.1 \pm (2.576) \times \frac{2.5}{\sqrt{100}}$$

$$= 6.1 \pm 2.576 \times \frac{2.5}{10}$$

$$= 6.1 \pm \frac{6.44}{10}$$

$$= 6.1 \pm 0.644$$

$$5.456 < \mu < 6.744$$



Q:- A wine manufacturer
 typical parameters
 $n=64$, $\bar{x}=9.0$

Sol:- Given that

1) Hypothesis:

$$H_0: \mu \leq 9.0$$

$$H_1: \mu > 9.0$$

2) Decide α -value

3) Choose the

Z-value

$$4) Z_{\alpha/2}$$

5) failed to reject

Q:- The EV

at most

water.

$$\alpha=0.01$$

$$n=121$$

$$1) H_0: \mu \leq 1$$

$$H_1: \mu > 1$$

$$2) \bar{x} - \mu$$

$$\frac{s}{\sqrt{n}}$$

$$= 1 - 1$$

$$\frac{6.33}{\sqrt{121}}$$

6 hours of
sample data

Q:- A wine manufacturer claims that his wine has atmost 9 ppm (parts per million) impurities in a barrel of wine.
 $n=64$, $\bar{x} = 9.06$ ppm, $s = 0.12$ ppm, Test at $\alpha = 0.01$.

SOL:- Given that

1) Hypothesis: Assume H_0 & H_1

$$H_0: \mu \leq 9 \text{ ppm}$$

$$H_1: \mu > 9 \text{ ppm}$$

2) divide α -value

$$\alpha = 0.01$$

3) choose the appropriate Test statistic

$$Z_{\text{statistic}} = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = 4$$

$$4) P(Z) = 0$$

$$\alpha = 0.01$$

$$P(Z) < \alpha$$

$$0 < 0.01$$

we have enough evidence to reject H_0

$$C.I. = \bar{x} \pm Z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

$$= 9.06 \pm 2.576 \cdot \frac{0.12}{\sqrt{64}}$$

$$\frac{9.06 - 9}{\frac{0.12}{\sqrt{64}}} = \frac{0.06}{\frac{0.03}{4}} = 8$$



Q:- The Evian water company claims that there are atmost 1 ppm units of wine in its highly overpriced water. Use the following sample data to test at $\alpha = 0.05$.
 $n=121$ bottles, $\bar{x} = 1.1$ ppm, $s = 0.33$ ppm.

$$1) H_0: \mu \leq 1$$

$$H_1: \mu > 1$$

$$2) \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$= \frac{1.1 - 1}{\frac{0.33}{\sqrt{121}}} = \frac{0.1}{\frac{0.33}{11}} = \frac{0.1}{0.03} = 3.03$$

$$\text{calculated } t = 3.03$$

$$\text{critical } t = 1.658$$

$3.03 > 1.658$, we reject null hypothesis

At the 5% significance level, there is strong evidence that the average wine content is greater than 1 ppm.

Two-Sample Z-Test

> Assumptions of two-sample z-test

> The sample sizes n_1 & n_2 of two samples drawn from two populations are large ($n_1 > 30$ & $n_2 > 30$)

> The samples are drawn from two normally distributed populations with corresponding standard deviation with σ_1 & σ_2 are known

$$Z_{\text{statistic}} = \frac{(\bar{X}_1 - \bar{X}_2) - (M_1 - M_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Q:- The dean of Saint Peter's school of management education believes that the graduating students with specialization in marketing earn at least 5000 more per month than the students with specialization in operations management to verify his belief. The dean collected a sample data from his graduating students. Conduct an appropriate test hypothesis test at $\alpha = 0.05$ to check whether the difference in monthly salary is at least 5000 more for students with marketing specialization compared to operation specialization. Assume that the salary of students with marketing specialization & operation specⁿ follow normal distribution.

Sample values on marketing & operations student.

Specialization	sample size	estimated mean salary	population standard deviation
marketing:	120 (n_1)	67,500 ($\bar{X}_1$)	7,200 (σ_1)
operations:	45 (n_2)	58,950 ($\bar{X}_2$)	4,600 (σ_2)

sd:- frame H_0 & H_1

$H_0: \mu_1 - \mu_2$

$H_1: \mu_1 - \mu_2$

2) decide α - v
 $\alpha = 0.05$

3) choose appropriate test
choose the

Z-test

4) P(3.

reje

Set frame H_0 & H_1

$$H_0: \mu_1 - \mu_2 \leq 5000$$

$$H_1: \mu_1 - \mu_2 > 5000$$

2) decide α value

$$\alpha = 0.05$$

3) choose appropriate test statistic

choose two-sample Z-test, because $n > 30$ in the question.

$$Z_{\text{statistic}} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$= \frac{(67500 - 58950) - 5000}{\sqrt{\frac{(7200)^2}{120} + \frac{(4600)^2}{45}}}$$

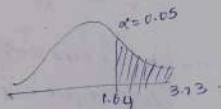
$$= \frac{3550}{\sqrt{432000 + 470222.22}}$$

$$= \frac{3550}{\sqrt{902222.22}} = \frac{3550}{949.85} = 3.73$$

$$4) P(3.73) = 1 - 0.999 = 0.001$$

$$P = 0.001 < \alpha = 0.05$$

reject null hypothesis (H_0).



Q:- Typing speed on a PC who types faster men/women.
test at $\alpha = 0.01$

men

$$\bar{x} = 65 \text{ wpm } (\bar{x}_1)$$

$$s = 10 \text{ wpm } (s_1)$$

$$n = 50 (n_1)$$

women

$$\bar{x} = 68 \text{ wpm } (\bar{x}_2)$$

$$s = 14 \text{ wpm } (s_2)$$

$$n = 60 (n_2)$$

as they didn't mention difference, consider $\mu_1 - \mu_2 = 0$

P:- Take homepay who earns more married/unmarried people

married

$$\bar{x} = 639.60 \$$$

$$s = 60 \$$$

$$n = 40$$

unmarried

$$\bar{x} = 658.20 \$$$

$$s = 90 \$$$

$$n = 60$$

Test at $\alpha = 0.04$

sol:-

$$1) \mu_1 - \mu_2 = 0$$

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

$$2) \alpha = 0.04$$

$$3) Z = \frac{639.60 - 658.20}{\sqrt{\frac{60^2}{40} + \frac{90^2}{60}}} = -1.24$$

$$4) -2.05 < -1.24 < 2.05$$

do not reject Null Hypothesis (H_0)

There is

Q:- who lives long

10-60 single women

$$\bar{x} = 78.5x$$

$$s = 140.4$$

$$n = 140$$

Q:- who has a

$$\bar{x} = 92$$

$$s = 12$$

$$n = 120$$

Q:- who lives longer married or unmarried women, test $\alpha = 0.01$

To do single women

$$\bar{X} = 78.5x$$

$$S = 140.4$$

$$n = 140$$

married women

$$77.04$$

$$16.04$$

$$160$$

$$p < \alpha$$

$$0 < 0.05$$

reject null hypothesis

↓
this true.

Q:- who has a longer life span. Test at $\alpha = 0.05$

non-retired

$$\bar{X} = 82.5 \text{ years}$$

$$s = 12 \text{ years}$$

$$n = 120$$

retired

$$72.5 \text{ years}$$

$$12.5 \text{ years}$$

$$n = 50$$

T-test

Both are bell-shaped
both follows normal distribution

z-test

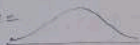
$n \geq 30$

known (σ)

> shape of distribution
is based on sample size

> tail part is very less

$z =$



t-test

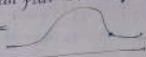
$n < 30$

unknown (σ)

> degree of freedom (sample size - 1)

> tail part is heavy tails.

$t =$



Student's t-distribution

> It is a member of continuous probability distributions

> T-test will be used when the sample size is small ($n < 30$)

& population standard deviation is unknown.

> It was developed by William Sealy Gossett under the pseudonym student.

> Distribution is equal to z-distribution

> The shape of the distribution depends on degree of freedom ($n-1$).

> If there are 'n' observations in the sample, and 'k' parameters are estimated from the sample then, degree of freedom is " $n-k$ ".

> The t-distribution is symmetric and bell-shaped like normal distribution but has heavier tails meaning that it is more prone to producing values that fall far from the mean.

For t-test,
$$C.I = \bar{X} \pm t_{(\alpha/2, OF)} \cdot \frac{s}{\sqrt{n}}$$

One sample t-test

Q:- A teacher claims that

70. A sample of 10

65, 68, 75, 70, 72, 74

level.

Sol:- 1) frame H_0 & H_1

$H_0: \mu = 70$

$H_1: \mu \neq 70$

2) decide $\alpha =$

$\alpha = 0.05$

3) choose ap

t statistic

$\bar{x} = 65$

Comput es

3) t statistic

4) determine

> decision

> significance

> From t

One sample t-test problems

Ex - A teacher claims that the average score of his students is 70. A sample of 10 students gave these scores 65, 68, 75, 70, 72, 69, 74, 71, 73, 67. Test at $\alpha = 0.05$ significance level.

Sol - 1) frame H_0 & H_1

$$H_0: \mu = 70$$

$$H_1: \mu \neq 70 \text{ (two-tailed test).}$$

2) decide α -value

$$\alpha = 0.05$$

3) choose appropriate test statistic.

$$t\text{-statistic} = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$$

$$\bar{X} = \frac{65 + 68 + 75 + 70 + 72 + 69 + 74 + 71 + 73 + 67}{10}$$

$$= \frac{704}{10} = 70.4$$

$$\text{Compute sample standard deviation } s = \sqrt{\frac{\sum (x_i - \bar{X})^2}{n-1}}$$

$$= 3.204$$

$$3) t\text{-statistic} = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$$

$$= \frac{70.4 - 70}{\frac{3.204}{\sqrt{10}}} = \frac{0.4}{\frac{3.204}{3.162}} = \frac{0.4}{1.013} \approx 0.3947$$

4) Determine the critical t-value

$$0.3947 > 0.05$$

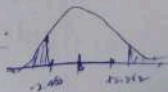
> degree of freedom (df) = $n-1 = 9$.

> significance level $\alpha = 0.05$

> From t-distribution table, $t_{0.025, 9} \approx 2.262$.

$$t_{(\frac{\alpha}{2}, df)}$$

We fail to reject the null hypothesis.



Q:- A manufacturer wants to test a hypothesis to see if the diameter of their 30m bolts are being manufactured properly. Test at $\alpha = 0.05$. He collected sample of 6 and their values are 30.02, 29.99, 30.11, 29.97, 30.01, 29.99.

Two sample t-test

> The test is used when the samples are independent. If you want to estimate difference in two populations means, then when standard deviation of the population are unknown.

2 types of Independent two-sample T-test, $\sigma_1^2 = \sigma_2^2 < 4$

1. Two sample t-test with equal variance (pooled variance)
This test is used only when it can be assumed that the two distributions have the same variance. $\sigma_1^2 = \sigma_2^2 \geq 4$.
2. Two sample t-test with unequal variance (Welch's t-test)
It is used only when the two population variances are not assumed to be equal.

Assumptions of 2-sample t-test with equal variance

- > Population variances σ_1^2 and σ_2^2 are unknown.
- > Population follows the normal distribution.
- > sample size is $(n_1 + n_2) > 40$.
- > Two samples are independent.
- > The two population variances are equal (i.e., $\sigma_1^2 = \sigma_2^2$).
- > degree of freedom is $n_1 + n_2 - 2$.

$$\text{std} = \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

formula for

Z-test

$$C.I. = \bar{X} \pm$$

$$Z_{\text{stat}} =$$

T-test

$$t_{\text{stat}} =$$

$$C.I. = \bar{X}$$

→ If σ^2

→ If σ^2

Q:- Suppose

tutorial

revealed

average

which

student

6 point

2 groups

to see
manufactured
6 and their
29.99.

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2} \quad S_p = \text{pooled variance}$$

formula for t-statistic $\Rightarrow t_{\text{statistic}} = \frac{(\bar{x}_1 - \bar{x}_2) - (M_1 - M_2)}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$

z-test

$$C.I. = \bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

$$z_{\text{stat}} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

T-test

$$t_{\text{stat}} = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$C.I. = \bar{x} \pm t_{(\alpha/2, DF)} \cdot \frac{s}{\sqrt{n}}$$

two-sample z-test

$$z_{\text{stat}} = \frac{(\bar{x}_1 - \bar{x}_2) - (M_1 - M_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

$$C.I. = (\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

two-sample t-test

$$t_{\text{stat}} = \frac{(\bar{x}_1 - \bar{x}_2) - (M_1 - M_2)}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$C.I. = (\bar{x}_1 - \bar{x}_2) \pm t_{(\alpha/2, df)} \sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$\Rightarrow$ If $\sigma_1^2 = \sigma_2^2 < 4$ (equal variance)

$\Rightarrow$ If $\sigma_1^2 = \sigma_2^2 \geq 4$ (unequal variance)

$\Rightarrow$ to identify
pooled or un-pooled
t-test.

Q:- Suppose we have 2 groups of students those who go for tutorial help and those who do not. A sample of 50 students revealed that 22 students had gone for tutorial help, the average grade for students who received tutorial help was 78 with a deviation of 5 points and the average grade for students who did not receive help was 72 with a deviation of 6 points. What is the 95% CI on the difference between these 2 groups of students. $\alpha = 0.05$.

$$n_1 = 22$$

$$\bar{x}_1 = 78$$

$$s_1 = 5$$

$$n_2 = 28 (50 - 22)$$

$$\bar{x}_2 = 72$$

$$s_2 = 6$$

$$S_p^2 = \frac{(21)(25) + (27)(36)}{22 + 28 - 2}$$

$$= \frac{525 + 972}{48}$$

$$= 31.1875$$

$$S_p^* = 972.66$$

1) frame H_0 & H_1

$$H_0: \mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 - \mu_2 \neq 0$$

2) Decide α -value

$$\alpha = 0.05$$

3) choose appropriate test statistic

$$s_p^* = 31.18$$

$$t_{\text{statistic}} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{31.18 \times \left(\frac{1}{33} + \frac{1}{33}\right)}}$$

$$= \frac{(78 - 72) - 0}{\sqrt{31.18 \times \left(\frac{28 + 22}{616}\right)}} = \frac{6}{\sqrt{2.530}} = \frac{6}{1.59} = 3.77$$

$$DOF = n_1 + n_2 - 2$$

$$= 50 - 2$$

$$= 48$$

$$\alpha = 0.05$$



Compare t -statistic with critical value

$$3.77 > 2.00$$

Reject Null Hypothesis.

∴ we want to test if there is a significant difference between the avg marks of

Boys: 72, 75, 78, 70, 74, 76

Girls: 68, 70, 65, 66, 69, 67

Test at $\alpha = 0.05$.

$$n_1 = 6 \quad n_2 = 6$$

$$\bar{x}_1 = 74.1, \quad \bar{x}_2 = 67.5$$

$$H_0: \mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 - \mu_2 \neq 0$$

$$s_p^2 = \frac{(n_1 - 1) \cdot s_1^2 + (n_2 - 1) \cdot s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{(5) s_1^2 + (5) s_2^2}{6 + 6 - 2}$$

$$s_1^2 = \frac{(74.1 - 72)^2 + (74.1 - 75)^2 + (74.1 - 75)^2 + (74.1 - 70)^2 + (74.1 - 74)^2 + (74.1 - 76)^2}{6 - 1}$$

$$s_1^2 = 2.85$$

$$s_2^2 = \frac{(67.5 - 68)^2 + (67.5 - 70)^2 + (67.5 - 69)^2 + (67.5 - 66)^2 + (67.5 - 69)^2 + (67.5 - 67)^2}{6 - 1}$$

$$= \frac{0.25 + 6.25 + 6.25 + 2.25 + 2.25 + 0.25}{5}$$

$$\sqrt{3.5} = 1.87$$

$$s_p^2 = \frac{5(2.85) + 5(3.5)}{10} = 3.175$$

$$5.81 \left(\frac{1}{6} + \frac{1}{6} \right)$$

$$= \frac{40.6 + 17.5}{10}$$

$$s_p^2 = 5.81$$

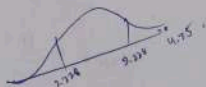
$$t_{\text{statistic}} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$= \frac{(74.1 - 67.5) - 0}{\sqrt{5.81 \left(\frac{1}{6} + \frac{1}{6} \right)}}$$

$$= 1.93$$

$$= \frac{6.6}{1.389} = 4.75$$

$$DOF = 12 - 2 = 10$$



$$\frac{50}{616} \approx 8.11\%$$

77

$$1/2 = 0.025$$

$$3.77$$

between

Q: A pharmaceutical company test a procedure for improving recovery score.

Treatment group: mean = 8.2, std = 2.4, $n_1 = 20$

Control group: mean = 6.9, std = 2.1, $n_2 = 22$

Test at $\alpha = 0.05$ and test whether there is a significant

sol difference in mean standard recovery score b/w treatment and control group.

sol 1) Frame null & Alternate hypothesis

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

$$2) \alpha = 0.05$$

3) mention $\Rightarrow$ if you are doing this test sample size ≥ 30 .

$$s_1^2 - s_2^2 < 4 \text{ (equal variance)}$$

$$t_{\text{stat}} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$S_p^2 = \frac{19(2.4^2) + 21(2.1^2)}{40}$$

$$S_p^2 = 5.05$$

$$t_{\text{stat}} = \frac{(1.3) - 0}{\sqrt{5.05 \left(\frac{1}{20} + \frac{1}{22} \right)}}$$

$$5.05 \left(\frac{22+20}{440} \right) \Rightarrow 5.05 \left(\frac{42}{440} \right) = \frac{212.1}{440} = 0.482$$

$$\sqrt{0.482} = 0.694$$

$$t_{\text{stat}} = \frac{1.3}{0.694} = 1.87$$

$$DOF = 42 - 2 = 40$$

$$DOF = 40$$

$$\alpha = 0.05$$

t-statistic value acceptance or rejection region.

So, failed to

$\therefore$ there is no

Q:- Two chips

chip x: mean

chip y: mean

Test at $\alpha = 0.05$

diff in me

equal

1) Frame Ho

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

$$2) \alpha = 0.05$$

$$3) S_p^2 =$$

t-stat

$x = 0.05$

Q:- Two chips are tested for performance score

chip 4: mean = 50, std = 6.5, n = 12.

1) Frame H_0 & H_1

$$H_0: \mu_1 = \mu_2$$
$$M_1 \neq M_2$$
 $\alpha = 0.05$

$$3) \quad s_p^2 = \frac{11(6)^2 + 11(6.5)^2}{22} = \frac{396 + 464.75}{22} = 39.125$$

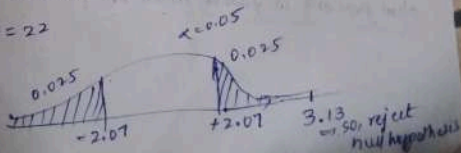
$$9p^2 = 39.125$$

$$t_{\text{stat}} = \frac{(58-50)(6)}{\sqrt{\frac{39.125}{6}}}$$

$$\frac{1}{12} + \frac{1}{12} = \frac{2}{12}$$

$$\sqrt{6.52}$$

$$= \frac{8}{2.55} \Rightarrow t_{stat} = 3.13$$

$$DOF = 22$$


If given is sample $\Rightarrow$ perform t-test
 If given is a population $\Rightarrow$ perform z-test.

There is a significant difference in performance.
 chip x performs better than y.

Q:- A company introduced a new delivery process

new process: $\bar{X}_1 = 120$, $std = 10$, $n_1 = 35$

old process: $\bar{X}_2 = 125$, $std = 10$, $n_2 = 35$

Test at $\alpha = 0.05$. Test whether the new process is faster.

Sol:- 1) Frame H_0 & H_1

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

$$\Rightarrow \alpha = 0.05$$

$$3) s_p^2 = \frac{34(100)}{68}$$

$$s_p^2 = \frac{34(100) + 34(100)}{68}$$

$$= \frac{6800}{68}$$

$$s_p^2 = 100$$

$$t_{\text{statistic}} = 2.09$$

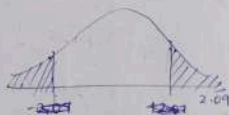
$$DOF = 70 - 2 = 68$$

$$-2.09 < 1.676$$

$t_{\text{statistic}}$ value is less than t-critical value.

So, we reject null hypothesis.

New process is faster than old process.



Two sample t

Assumptions:-

1)

2) The population

3) The two sample

4) Two sample

5) The sample

means ($\bar{X}_1$,

and standard

degree of

$t_{\alpha/2}$

Two sample t-test with unequal variance.

Assumptions:-

- 1)
- 2) The population follows normal distribution are $n_1 + n_2 < 40$
- 3) The two samples are independent.
- 4) Two sample variances are not equal. ($\sigma_1^2 \neq \sigma_2^2$)
- 5) The sampling distribution of the difference in estimate of means ($\bar{x}_1 - \bar{x}_2$) follows t-distribution with mean $\mu_1 - \mu_2$ and standard deviation is $s_{\mu} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

degree of freedom is
$$\left[\frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{\left(\frac{s_1^2}{n_1} \right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2} \right)^2}{n_2 - 1}} \right]$$

$$t_s = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Sol:-

$$n_1 = 6$$

$$n_2 = 6$$

$$\bar{x}_1 = 74.16$$

$$\bar{x}_2 = 67.5$$

$$\alpha = 0.05$$

$$s_1 = 2.8586$$

$$s_2 = 1.87$$

$$H_0: \mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 - \mu_2 \neq 0$$

$$S_{\mu} = \sqrt{\frac{(2.85)^2}{6} + \frac{(1.87)^2}{6}}$$

$$= \sqrt{\frac{8.1225}{6} + \frac{3.4969}{6}}$$

$$= \sqrt{1.3537 + 0.5828} = 1.7492$$

$$S_{\mu} = 1.39$$

$$t_3 = 4.78$$

$$DOF = 8$$

$$\alpha = 0.05$$

$$DOF = (S_{\mu})^4$$

$$(1.7492)^4$$

$$\frac{\left(\frac{s_1^2}{n_1}\right) + \left(\frac{s_2^2}{n_2}\right)}{\frac{n_1-1}{n_1-1} + \frac{n_2-1}{n_2-1}}$$

$$\frac{\left(\frac{2.85^2}{6}\right) + \left(\frac{1.87^2}{6}\right)}{\frac{5}{5} + \frac{5}{5}}$$

$$= \frac{3.733}{\frac{(1.35)^2}{5} + \frac{(0.58)^2}{5}}$$

$$= \frac{3.733}{0.36 + 0.067} = \frac{3.733}{0.427}$$

$$= \frac{3.733}{0.427}$$

$$DOF = 8.74$$



$$4.78 > 2.306$$

test falls in rejection area. so we reject null hypothesis.

Q:- A researcher is plant growth plants growth plants growth The data is a

group 1	21	16
group 2	23	2

can the res can if that is a plants b/w

Q:- A researcher is investing the effect of new fertilizer on a plant growth. They measure the heights of 10 random plants growth with the new fertilizer (group 1) and 12 random plants growth with a standard fertilizer (group 2).

The data is as follows

group 1	21	16	22	23	19	24	20	25	17	17		
group 2	23	23	22	21	26	23	22	22	24	20	21	24

can the researcher use when 2 sample t-test to determine if there is a significant difference in the average height of plants b/w the two groups. Test at $\alpha = 0.10$.

Q- A company wants to compare customer satisfaction scores b/w two different stores. $\alpha = 0.05$

Store A
 $\bar{x}_1 = 80$
 std = 6
 $n = 40$

Store B
 $\bar{x}_2 = 75$
 std = 9
 $n = 35$

DOF = 57

Test if there is a significant difference in customer satisfaction b/w two stores.

Soln- $H_0: \mu_1 - \mu_2 = 0$

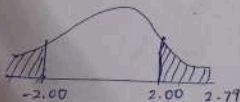
$\alpha = 0.05$

$H_1: \mu_1 - \mu_2 \neq 0$

$$S_{SE} = \sqrt{\frac{36}{40} + \frac{81}{35}} = \sqrt{0.9 + 2.314} = \sqrt{3.214} = 1.7$$

$$t_{\text{stat}} = \frac{80 - 75 - (0)}{1.7} = 2.79$$

$$\begin{aligned} \text{DOF} &= \frac{(3.214)^2}{\frac{(0.9)^2}{39} + \frac{(2.314)^2}{34}} = \frac{10.329}{\frac{0.81}{39} + \frac{5.354}{34}} \\ &= \frac{10.329}{0.02 + 0.15} = \frac{10.329}{0.177} = 58.3 \end{aligned}$$



$t_{\text{stat}} > CV$

$\therefore$ we reject null hypothesis

$\therefore$ Therefore, there is a significant difference in customer satisfaction b/w two stores.

30012

Pooled variance problem

8. A company claims that children (age 7-12) who drink their health drink will grow taller than the children who do not drink that health drink.

Given table show avg increase in height over a year period.

$\alpha = 0.05$ Test whether the increase in height for the children who drink health drink is atleast 1.5 cm

action

<u>Group</u>	<u>size</u>	<u>Avg height (in cm)</u>	<u>S.S.D.</u>
Drink health drink	80	7.6 cm	1.1 cm
do not drink health drink	80	6.3 cm	1.3 cm

Paired T-Test (dependent variables)

This test is used when the samples are dependent i.e., when there is only one sample that has been tested twice (or) when there are two samples that have been matched or paired.

Examples:-

- 1) Sugar levels before and after eating
- 2) Heart rate before drinking a cup of coffee and after.
- 3) Alcohol consumption before attending course and after completion of course and after marriage.
- 4) Maths scores before attending course and after completion of course.
- 5) Test scores based on age groups.

$$t_0 = \frac{\bar{x}_d - \mu_d}{\frac{s_d}{\sqrt{n}}}$$

$\bar{x}_d$ - sample mean difference b/w two samples

s_d - standard deviation difference b/w two samples

μ_d - population mean difference

$\Rightarrow \bar{x}_d$ follows normal distribution.

• A paired t-test is used when the means of two groups are being compared, and the groups are dependent (i.e., the same group is being compared at two different times).

• Paired test is used when the samples are dependent (i.e., the same group is being compared at two different times).

(i) the same treatment

(ii) matched pairs

(iii) when the data is paired

(iv) The data is paired

Q:- A company has records of

Test at $\alpha = 0.05$

Before

After

sol:- Diff

(i) $H_0: \mu_d = 0$

$H_1: \mu_d \neq 0$

(ii) $\alpha = 0.05$

(iii) $t_0 =$

ndent i.e., when
d twice (or)
matched or

• A paired t-test is a statistical test used to compare the means of two related groups to see if there is a significant difference b/w them.

• Paired test is applied when

(i) the same subjects are measured before and after a treatment

(ii) matched subjects are used (pairs)

(iii) when data is continuous.

(iv) the differences are approximately normally distributed.

Q - A company gives 6 employees a training program and records of their test scores before and after training.

Test at $\alpha = 0.05$

Before	50	60	55	58	62	53
After	55	63	54	60	65	57
	↓	↓	↓	↓	↓	↓
sol:- Diff:	5	3	-1	2	3	4

• (i) $H_0: \mu_1 = \mu_2$ $\mu_d = 0$
 $H_1: \mu_1 \neq \mu_2$ $\mu_d \neq 0$

(ii) $\alpha = 0.05$

(iii) $t_s = \frac{\bar{x}_d - \mu_d}{\frac{s_d}{\sqrt{n}}}$

$$\bar{x}_d = \frac{5+3-1+2+3+4}{6} = 2.666$$

$$\bar{x}_d = 2.67$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$s_d = \sqrt{\frac{\sum (x_i - \bar{x}_d)^2}{n-1}}$$

$$s_d = \sqrt{\frac{(5-2.67)^2 + (3-2.67)^2 + (-1-2.67)^2 + (2-2.67)^2 + (3-2.67)^2 + (14-2.67)^2}{5}}$$

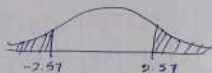
$$s_d = 5.4289 + 0.1089 + 13.4689 + 0.4489 + 0.1089 + 11.7689$$

$$= \sqrt{24.266}$$

$$= 2.07$$

$$t_s = \frac{\bar{x}_d - M_d}{s_d/\sqrt{n}} = 3.16$$

$$Dof = n-1 = 5$$



$$c.v = 2.57$$

$$t_s > c.v$$

$$3.16 > 2.57$$

$\therefore$ So, reject null hypothesis.

t_{stat} falls in rejection region.

Q- Suppose 5 people's weights are measured before and after a one month diet

$$\alpha = 0.05$$

Before	80	75	90	85	95
after	78	74	85	83	92
diff	2	1	5	2	3

$$\frac{13}{5} = 2.6$$

$$\bar{x}_d = 2.6$$

$$s_d = \sqrt{\frac{(2-2.6)^2 + (1-2.6)^2 + (5-2.6)^2 + (2-2.6)^2 + (3-2.6)^2}{4}}$$

$$= \sqrt{0.36 + 2.56 + 11.56 + 0.36 + 0.16}$$

$$t_s = \frac{\bar{x}_d - M_d}{s_d/\sqrt{n}}$$

$$Dof = n-1$$

$$= 4$$

$$t_s > c.v$$

$$3.801 > 2.77$$

$\therefore$ we reject

Q- A researcher

on Mondays the

sample of 50

estimated as

5.5 ml. conduct

To check the

more coffee

week.

3d- 1) $H_0: M_d$

$H_1: M_d$

$$2) \bar{x}_d = 1$$

$$t_s =$$

$$s_d = \sqrt{\frac{(2-2.6)^2 + (1-2.6)^2 + (3-2.6)^2 + (2-2.6)^2 + (3-2.6)^2}{4}}$$

$$= \sqrt{\frac{0.36 + 2.56 + 0.16 + 0.36 + 0.36}{4}} = \sqrt{2.35} = 1.53$$

$$t_s = \frac{\bar{x}_d - \mu_d}{s_d/\sqrt{n}} = \frac{2.6 - 0}{\frac{1.53}{\sqrt{5}}} = \frac{2.6}{0.684} = 3.801$$

$$DOF = n - 1$$

$$= 4$$

$$t_s > t_{\alpha/2}$$

$$3.801 > 2.77$$

$\therefore$ we reject null hypothesis



Q. A researcher believes that people drink more coffee on Mondays than other days of the week. Based on the sample of 50 coffee drinkers the mean difference was estimated as 14 mL. The corresponding std deviation was 8.5 mL. conduct an appropriate hypothesis test at $\alpha = 0.1$. To check the claim that people drink an avg of 10 mL more coffee on Mondays, compared to other days of the week.

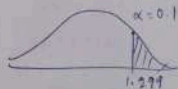
$$H_0: \mu_d \leq 10$$

$$H_1: \mu_d > 10$$

$$\mu_d = 10, n = 50,$$

$$\alpha = 0.1, s_d = 8.5$$

$$\bar{x}_d = 14$$



$$t_s = \frac{\bar{x}_d - \mu_d}{s_d/\sqrt{n}} = \frac{14 - 10}{8.5/\sqrt{50}} = \frac{4}{1.202} = 3.327$$

$$DOF = 50 - 1 = 49$$

Hypothesis Testing for population proportion.

→ One-sample Test for proportion.

According to the central-limit theorem of proportions, the sampling distribution of proportion ($\hat{p}$ or $\bar{p}$) for a large sample follows an approximate normal distribution with mean (p) and standard deviation $\sqrt{\frac{p(1-p)}{n}}$ that is the z statistics as defined below will follow the std normal distribution.

$$\text{Formula } z_{\text{proportion}} = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

$\hat{p} = 0.12$

1. According to a study exactly 12% of gift cards purchased from e-commerce portals are never used. The manager of an e-commerce company wanted to test whether this claim is true?

She collected data of 250 gift cards purchases and found that 22 gift cards were not used till its expiry date.

(i) conduct an appropriate hypothesis test at 5% significance to check whether the claim that exactly 12% gift cards are never used is true or not.

(ii) Calculate the 95% confidence interval for the proportion of gift cards that are not used.

Sol: (i) $H_0: p = 0.12$

$H_1: p \neq 0.12$

(ii) $\alpha = 0.05$

$$(iii) \hat{p} = \frac{22}{250} = 0.088$$

$n = 250$

$$z_s = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

$$z_s = -1.55$$



z_s value is not

So, we failed to

exactly 12% gift

mean $\rightarrow$ CI

proportion CI

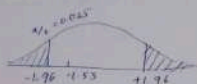
The proportion

and upper

portions, the
large sample
mean (μ) and
as defined

$$Z_s = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} = \frac{0.088 - 0.12}{\sqrt{\frac{0.12(1-0.12)}{250}}} = \frac{-0.032}{0.017} = -1.55$$

$$Z_s = -1.55$$



Z_s value is not the part of rejection region
So, we failed to reject null hypothesis.
Exactly, 12% gift cards are never used is true.

urchased from
n e-commerce
ue Z

d found of

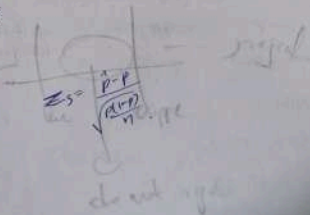
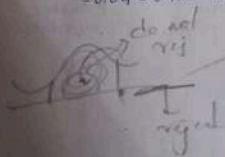
ificance
cards

on of gift

$$\begin{aligned} \text{mean} &\rightarrow CI = \bar{x} \pm Z_{\alpha/2} \frac{s}{\sqrt{n}} \\ \text{proportion } CI &= \hat{p} \pm Z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}} \Rightarrow [-0.04 \text{ to } 0.2] \\ &= 0.088 \pm 1.96 \sqrt{\frac{0.12(0.88)}{22}} \\ &= 0.088 \pm 0.135 \\ &= -0.04 \text{ to } 0.223 \end{aligned}$$

The proportion of unused gift cards lie b/w lowerbound and upperbound. So, we do not reject null hypothesis.

$$-0.04 < 0.12 < 0.223$$



Two Sample proportion test

When the proportions are estimated from a large sample then sampling distribution of the proportions follows a normal distribution according to the central limit theorem.

Let $\hat{p}_1$ & $\hat{p}_2$ be the estimated values of proportions from large samples. The difference $p_1 - p_2$ is the hypothesized difference in population proportions.

mean :- $Z_s = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$

proportion :- $Z_s = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\hat{p}(1-\hat{p}) \times \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$

sample population
 $\hat{p}_1 - \hat{p}_2$
 $\left(\frac{x_1}{n_1} - \frac{x_2}{n_2}\right) - \left(\frac{\mu_1}{n_1} - \frac{\mu_2}{n_2}\right)$

$\hat{p}$ = pooled estimation for the proportion.

1. The marketing manager of the company believes that the non-affluent customers are sensitive to discounts compared to affluent customers. To validate this hypothesis discount coupons was sent to non-affluent & affluent customers, and the data is given below. Use an appropriate hypothesis to check whether there is any difference in proportion of customers who use discount coupons at $\alpha = 0.05$

Group	Sample size	No. of customers using discount coupons	Estimated proportion
non-affluent	500 (n_1)	145	0.29
affluent	300 (n_2)	42	0.14

(i) $H_0: p_1 = p_2$

(ii) $\alpha = 0.05$

$H_1: p_1 \neq p_2$

$$\hat{p} = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2}$$

Here, $\hat{p}$ means pooled estimation for the proportion.

$$\hat{p}_1 = \frac{145}{500}$$

$$= 0.29$$

$$\hat{p} = \frac{(500)(0.29) + (300)(0.14)}{500 + 300}$$

$$Z_s = 4.1$$

Parametric Test

1) It makes a parametric distribution normal distribution

2)

3) It requires data means measurable

3) Example is

t-test, Z-test, Pearson correlation

4) Efficiency when assumptions are met

5) Sample size parametric

6) distribution

7) We need

9) mean

$$Z_s =$$

if sample then
a normal

ns from large
d difference

$$\hat{p}_1 = \frac{195}{500}$$

$$= 0.39$$

$$\hat{p}_2 = \frac{42}{300}$$

$$= 0.14$$

$$\hat{p} = \frac{(500)(0.39) + (300)(0.14)}{500 + 300} = 0.235$$

$$Z_s = 4.95$$

Parametric Test

1) It makes assumptions about parametric distribution means normal distribution

2)

3) It requires interval or ratio data means continuous and measurable data

4) Examples for parametric test, t-test, z-test and anova, Pearson correlation

5) Efficiency is more powerful when assumptions are met

6) Sample size is larger in parametric test

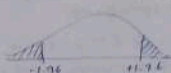
7) Distribution is symmetric

8) We need summary statistics

eg - mean, std deviation

$$Z_s = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$\hat{p} = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2}$$



$Z_s > \text{Z critical value}$

Non-parametric Test

1) It makes no assumptions about non-parametric test. Non-parametric test is a distribution free test

2) It works with ordinal or nominal data (ranks categories, satisfaction levels and genders)

3) Examples - chi-square test

4) Efficiency is less powerful but more flexible when assumptions are violated

5) sample size is smaller

6) distribution is not symmetric

7) We don't need summary statistics.

It is not about the population parameters.

$$\frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

(*) Chi-Square test

It is used to determine the relationship between categorical variables. It is a non-symmetric. chi-square distribution will have the positive values (0 to ∞).

chi-square distribution depends on degree of freedom.

If degree of freedom changes distribution also changes.

chi-square is a non-parametric test

chi-square test $\left\{ \begin{array}{l} \text{chi-square test for goodness of fit test} \\ \text{chi-square test for independence test} \end{array} \right.$

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

→ observed values
→ expected values

O_i means observed frequency in category

E_i means expected frequency in category

$$\text{Dof} = n - 1$$

↓
no. of categories

Conclusion:- If $\chi^2 > \text{critical value} \Rightarrow \text{reject } H_0$

$\chi^2 < \text{critical value} \Rightarrow \text{fail to reject } H_0$

$p < \alpha \Rightarrow \text{reject}$

Q:- Hanuman Airlines operated daily flights through several Indian cities. one of the problems hanuman airlines faces is food preferences by the passengers. Captain took the operations manager of HA believes that 35% of their passengers prefer vegetarian food and 5% request for diabetic food. A sample of 500 passengers was chosen to analyse the food preferences and the data is shown in table.

Conduct a chi-square test to see if the belief is true at a 5% level of significance.

Food type:

No. of passengers

H_0 : Probability of

H_1 :

$\alpha = 0.05$

observed values

Expected values

$O_i - E_i$

$\frac{(O_i - E_i)^2}{E_i}$

$\chi^2 =$

Dof = 4 - 1 = 3

Conduct a chi-square test to check whether the captain's cook's belief is true at $\alpha = 0.05$

Food type:	Vegetarian	Non-vegetarian	Low calorie	Diabetic food
No. of passengers	170	185	90	35

H_0 : Probability distribution of food preferences are equal.
 H_1 : " " " " " " " " not equal

2) $\alpha = 0.05$

Observed values	170	185	90	35
Expected values	500×0.35 <u>175</u>	$500 \times \frac{40}{100}$ <u>200</u>	$500 \times \frac{20}{100}$ <u>100</u>	$500 \times \frac{5}{100}$ <u>25</u>
$O_i - E_i$	15	-15	-10	10
$\frac{(O_i - E_i)^2}{E_i}$	$\frac{15^2}{175}$ <u>1.28</u>	$\frac{(-15)^2}{200}$ <u>1.125</u>	$\frac{(-10)^2}{100}$ <u>1</u>	$\frac{10^2}{25}$ <u>4</u>

Sum = 7.405

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} = 7.405$$

$$DOF = 4 - 1 = 3$$

$$C.V. = 7.815$$

$$\chi^2 < C.V.$$

$$7.405 < 7.815$$

fail to reject

Q:- A dice was rolled 120 times observed face counts
18, 22, 20, 16, 24, 20. Test at $\alpha = 0.05$.

set:- H_0 : Dice is fair

H_1 : Dice is not fair.

observed	18	22	20	16	24	20
Expected	20	20	20	20	20	20
$O_i - E_i$	-2	2	0	-4	4	0
$\frac{(O_i - E_i)^2}{E_i}$	$\frac{(-2)^2}{20}$	$\frac{2^2}{20}$	$\frac{0^2}{20}$	$\frac{(-4)^2}{20}$	$\frac{4^2}{20}$	$\frac{0^2}{20}$
	0.2	0.2	0	0.8	0.8	0

$$\chi^2 = 2$$

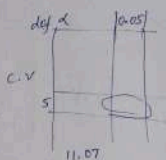
$$DOF = n - 1$$

$$= 6 - 1 = 5$$

$$\chi^2 < C.V$$

failed to reject null hypothesis.

so, dice is fair.



chi-square test

Independent test

It is a non-parametric test

your data is not

It determines whether two

independent variables

If two variables are

or certain values

Expected frequency

$$\chi^2 =$$

$$df = (r - 1)$$

Independent variables.

Q:- A movie

relationship

they buy a

goes. Test

observed data

movie type

Action

comedy

Total

H_0 : The

H_1 : The

Q:- A candy company claims that its bags of mixed fruit juice
chews contain an equal no. of 5 flavours - apple, lime, cherry,
orange & grape. A random sample of 10 bags is taken and
the no. of each flavour is counted. the results are shown in
table at 5% significance level. Does the data provide evidence
that the flavour distribution is different from the company's
claim.

flavour	apple	lime	cherry	orange	grape
Observed count	180	250	120	225	225

H_0 : The proportion of each flavour in the population of candy
bags is equal.

H_1 : ... not equal

Chi-square test for Independence

(Independent test / test of association).

It is a non-parametric test. It is used to determine whether your data is significantly different from what you expected. It determines whether two categorical variables related or not. (dependent / independent).

If two variables are related, probability of one variable having a certain value is dependent on the value of the other variable.

$$\text{Expected frequency } (E_i) = \frac{\text{Row total} \times \text{column total}}{N}$$

N
↓
Grand total.

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

$$df = (r-1) \times (c-1)$$

$r \rightarrow \text{row}$
 $c \rightarrow \text{column}$

Independence test is to test association between 2 categorical variables.

Q:- A movie theatre manager wants to know if there is a relationship b/w the type of movie people watch and whether they buy a snack. The manager collects data from 200 movie goers. Test at $\alpha = 0.05$

observed data

movie type	snacks	no-snacks	Total
Action	60	40	100
comedy	30	70	100
Total	90	110	$N=200 \rightarrow \text{grand total.}$

H_0 :- The two variables are independent (There is no association)

H_1 :- The two variables are dependent (There is an association)

$$T_{1,1}(60) = \frac{100 \times 70}{200} = 45$$

$$T_{1,2}(40) = \frac{100 \times 110}{200} = 55$$

$$T_{2,1}(91) = \frac{100 \times 70}{200} = 45$$

$$T_{2,2}(70) = \frac{100 \times 110}{200} = 55$$

Expected values.

finding chi-square statistic value

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$\chi^2 = \frac{(60-45)^2}{45} + \frac{(40-55)^2}{55} + \frac{(90-45)^2}{45} + \frac{(70-55)^2}{55}$$

$$\chi^2 = 18.18$$

$$DOF = (2-1) \times (2-1) = 1$$

critical value is 3.841

$$\alpha = 0.05$$

$$\chi^2 > \text{critical value.}$$

Reject null hypothesis. (H_0 is true) The variables are dependent.

Q- The table below shows the avg no. of hours students spend studying for class everyday in a high school. Is the avg no. of hrs dependent on the type of student. Test at $\alpha = 0.05$

student	0-2hrs	2-4hrs	4-6hrs	
Juniors	76	143	91	310
Seniors	147	109	64	320
	223	252	155	630

3d-
H₀: The avg no. of hrs is same

H₁:

Expected values

$$T_{1,1}(76) = \frac{310 \times 223}{630}$$

$$T_{1,2}(143) = \frac{310 \times 252}{630}$$

$$T_{2,1}(91) = \frac{320 \times 223}{630}$$

finding the chi-square

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$= \frac{(110-76)^2}{110} + \dots$$

$$= \frac{(128-109)^2}{128} + \dots$$

$$= \frac{115.6}{110} + \frac{3}{1}$$

$$= 10.5 + 2$$

$$= 32.26$$

$$DOF =$$

Reject

H₀

14. The no. avg no. of hours students spend for studying is same. $\therefore$ is not same.

Expected values

$$T_{11} E_1 (74) = \frac{310 \times 320}{630} = 157.76110$$

$$T_{21} E_1 = \frac{228 \times 320}{630} = 113$$

$$T_{12} E_2 (143) = \frac{310 \times 252}{630} = 124$$

$$T_{22} E_2 = \frac{252 \times 320}{630} = 128$$

$$T_{13} E_3 (79) = \frac{310 \times 155}{630} = 76$$

$$T_{23} E_3 = \frac{155 \times 320}{630} = 77$$

finding the chi-square statistic value,

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

$$= \frac{(110-76)^2}{76} + \frac{(124-143)^2}{143} + \frac{(76-71)^2}{71} + \frac{(113-147)^2}{147}$$

$$+ \frac{(128-109)^2}{109} + \frac{(79-64)^2}{64}$$

$$= \frac{1156}{76} + \frac{361}{143} + \frac{25}{71} + \frac{1156}{147} + \frac{361}{109} + \frac{225}{64}$$

$$= 10.5 + 2.91 + 2.96 + 10.23 + 2.82 + 2.84$$

$$= 32.26$$

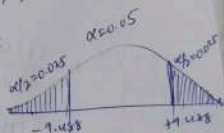
$$DOF = (3-1) \times (3-1) = 2 \times 2 = 4$$

$$\chi^2 > \text{critical value}$$

$$32.26 > 9.488$$

Reject null hypothesis

H_0 is true



Q- 500 elementary school boys and girls are asked which is their favourite colour: blue, green or pink.

	Blue	Green	Pink	
Boys	100	150	20	270
Girl	20	30	180	230
	120	180	200	500

Test at $\alpha = 0.05$. would you conclude that there is a relationship between gender and favourite colour.

H₀: There is no relationship b/w gender & fav colour

H₁: There is a relationship " " " " "

$$\alpha = 0.05$$

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

$$\tau_{11} = \frac{270 \times 120}{500} = 64.8$$

$$\tau_{12} = \frac{270 \times 180}{500} = 97.2$$

$$\tau_{13} = \frac{270 \times 200}{500} = 108$$

$$\tau_{21} = \frac{230 \times 120}{500} = 55.2$$

$$\tau_{22} = \frac{230 \times 180}{500} = 82.8$$

$$\tau_{23} = \frac{230 \times 200}{500} = 92$$

$$= \frac{(100-64.8)^2}{64.8} + \frac{(150-97.2)^2}{97.2} + \frac{(20-108)^2}{108} + \frac{(20-55.2)^2}{55.2} +$$

$$\frac{(30-82.8)^2}{82.8} + \frac{(180-92)^2}{92}$$

$$\frac{1239.04}{64.8} + \frac{2187.84}{97.2} + \frac{7744}{108} + \frac{1239.04}{55.2} + \frac{2187.84}{82.8}$$

$$+ \frac{7744}{92} = 19.12 + 28.68 +$$

F-Test

- Comparing two variables
Z-Test and t-test

(H₀, H₁)

F-test is used to check same variance or not

One which has less more variance is better

F-test is developed by

F-distribution was

from the two independent random samples

F-distribution is

Here, M₁, M₂ are

are degrees of freedom

F-distribution is

$$X = \frac{U_1}{U_2}$$

then the probability

$$f(x) =$$

using the a

we do not

Instead of

The random

$$X =$$

F-Test

- Comparing two variables.

$\bar{x}$ -Test and t -test are used to compare two mean values

(μ_1, μ_2) .

F-test is used to check two normal populations ^{variables} have same variance or not.

One which has less variance is good work and one which has more variance is bad work.

F-test is developed by Ronald Fisher.

F-distribution was to study the behaviour of two variances from the two independent normal population random sample taken from.

F-distribution is skewed to right side. It is not a symmetric

Here, u_1, u_2 are chi-square random variables with D_1 and D_2 are degrees of freedom, u_1 and u_2 are independent variables.

F-distribution is always positive (or) $x \geq 0$.

$$X = \frac{u_1/D_1}{u_2/D_2}$$

then the probability density function for x is given by

$$f(x) = \frac{\gamma\left(\frac{D_1+D_2}{2}\right)\left(\frac{D_1}{D_2}\right)^{D_1/2}}{\gamma\left(\frac{D_1}{2}\right)\gamma\left(\frac{D_2}{2}\right)} \times \frac{x^{D_1/2-1}}{\left(1+\frac{D_1}{D_2}x\right)^{\gamma\left(\frac{D_1+D_2}{2}\right)}}, x \geq 0$$

using the above formula, the critical value is found but we do not integrate the complex functions.

Instead of this we use F-table.

The random variable of the F-distribution may also be written as

$$X = \frac{S_1^2}{\sigma_1^2} \div \frac{S_2^2}{\sigma_2^2} \quad \text{where,}$$

F-distribution is non-symmetric and shape of the distribution depends on the values of d_1 and d_2 ($d_1, d_2 =$

F-distribution is used in analysis of variance (ANOVA) test to test mean values of multiple groups.

F-test can be used to know - two samples come from populations with equal variances or different variances.

for dataset

Q:- The dataset contains 480 Branic strength measurements for two batches of material. The summary statistics of each batch are shown below

Batch 1

$n_1 = 240$
mean = 688.99
 $s_1 = 65.5$

Batch 2

$n_2 = 240$
mean = 611.15
 $s_2 = 61.8$

Test the variances for the two batches are equal. Test at $\alpha = 0.05$

Sol:- 1) Frame H_0 & H_1

$H_0: \sigma_1^2 = \sigma_2^2$

$H_1: \sigma_1^2 \neq \sigma_2^2$

2) $\alpha = 0.05$

3) $\Rightarrow$ choose an appropriate test

$$F_{\text{statistic}} = \frac{\text{variance 1}}{\text{variance 2}} = \frac{s_1^2}{s_2^2}$$

$$F_{\text{statistic}} = \frac{4290.25}{3619.24}$$

$$= 1.183$$

$$DOF \Rightarrow [d_1 = n_1 - 1] [d_2 = n_2 - 1]$$

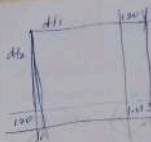
$$d_1 = 239, d_2 = 239$$

$$\alpha/2 = 0.025 \text{ (0.025 table)}$$

for critical

Optimal $F_{(1-\alpha/2, d_1, d_2)}$

$$\text{left tail} = \frac{1}{\text{Optimal value}} \Rightarrow \frac{1}{F_{(1-\alpha/2, d_1, d_2)}}$$



If $F_{\text{statistic}} < \text{critical value}$
we got $F_{\text{statistic}} < \text{critical value}$

Fail to reject H_0

Q:- Is there a difference in the best selling 15 new york times books in 2017 weeks had a std of 0.17 weeks can we conclude Test at $\alpha = 0.05$

$$n_1 = 15$$

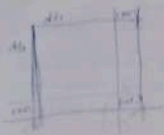
$$\sigma = 0.17$$

H_0 : the variance is equal

H_1 : The variance is not equal

This is a two-tailed test

F-statistic



$\text{Test Statistic} = \text{critical value} \rightarrow \text{reject null hypothesis}$
 $\text{weight} = \text{F statistic} = 1.123$
 $\text{critical value} = 1.14$
 $\text{F statistic} < \text{c.v.}$
 $1.123 < 1.14$
 Fail to reject H_0

Q: Is there a difference b/w variances of the no. of weeks on the best seller list for non-fiction and fiction books
 15 newyork times best selling fiction books had a std of 6.17 weeks on the list. 18 newyork times best non-fiction had a std of 13.12 weeks at 10% significance level.
 Can we conclude there is a difference in the variance.
 Test at $\alpha = 0.05$

$n_1 = 15$ $n_2 = 18$
 $s_1 = 6.17$ $s_2 = 13.12$

conclusion
 $F_5 > c.v$
 $4.52 > 3.29$
 Reject H_0

H_0 : The variances are equal, $\sigma_1^2 = \sigma_2^2$
 H_1 : The variances are not equal, $\sigma_1^2 \neq \sigma_2^2$
 This is a two-tailed test

$$\begin{aligned}
 F_{\text{STATISTIC}} &= \frac{s_1^2}{s_2^2} \quad \left(\begin{array}{l} \text{larger variance} \\ \text{in numerator} \end{array} \right) \\
 &= \frac{(6.17)^2}{(13.12)^2} = 4.52 \\
 &\text{Since } s_2^2 > s_1^2,
 \end{aligned}$$

$s_1^2, s_2^2 \rightarrow$ sample variances
 * larger variance should be placed in the numerator to make sure $F \geq 1$.

$$\begin{aligned}
 F_5 &= \frac{172.13}{38.07} = 4.52 \\
 d_1 &= 15, \quad d_2 = 14 \\
 &\quad \downarrow \quad \quad \downarrow \\
 &\text{numerator} \quad \text{denominator}
 \end{aligned}$$

$$\begin{aligned}
 F_{0.025, 15, 14} &\approx 3.29 \\
 F_{0.975, 15, 14} &= \frac{1}{3.44} \\
 &= 0.29
 \end{aligned}$$

ANOVA: Analysis of Variance

one-way Anova
Two-way Anova

↓
Anova test is a statistical test used to compare the means of three or more independent groups to see if at least one group mean is significantly different from the others.

Why do we need ANOVA?

It is used as an alternative to the multiple t-test, if we want to compare three groups we could do multiple t-test/group A vs group B and group B vs group C. But this will increase the chance of type-I error.

Anova test is used to test all groups at once while keeping the error rate under control.

Types of Anova

One-way Anova

Two-way Anova

- > One-way Anova is a hypothesis test in which only single factor taken into consideration.
- > Two-way Anova examines the effect of two independent factors.

One-way Anova

- It is a statistical test that can be used to compare means of two or more samples.
- This technique can be used only for one-factor (one independent variable), i.e., one-way).

If we want to study the effect of 3 levels ($\alpha_1, \alpha_2, \alpha_3$) of a fertilizer on plant growth where $\alpha_1, \alpha_2, \alpha_3$ are the 3 levels of the factor being studied.

Assumptions of Anova

1) Independence of data
The data collected
Example:- Scores & influence

2) Normality:-

The data in each distributed.

3) Homogeneity of

The variance in

$$\sigma_1^2 = \sigma_2^2 = \sigma_3^2$$

Steps to solve t

1) Frame H_0 & H_1

H_0 : All group

H_1 : At least one

2) Choose the

3) Calculate

4) Compute the

SST: Total

* SSB: Sum

of

for

* SSW: Sum

5) Compute

dfB

Assumptions of Anova Test

1) Independence of observations:-

The data collected from different groups should be independent.

example:- Scores of students in one class should not influence scores of another class.

2) Normality:-

The data in each group should be approximately normally distributed.

3) Homogeneity of variance:- (Equal variance).

The variance in each group should be the same i.e.,

$$\sigma_1^2 = \sigma_2^2 = \sigma_3^2$$

Steps to solve the Anova Test

If question more than 2 means

1) then perform Anova test

1) Frame the H_0 & H_1

H_0 : All group means are equal

H_1 : At least one group mean is different

2) choose the significance level (α value)

3) calculate group means and the overall mean

4) compute the sum of squares

SST:- Total sum of squares $\rightarrow SST = \sum (X_i - \bar{X})^2 = SST + SSB + SSW$

* SSB: Sum of squares for between groups

$$\text{formula} \rightarrow \sum n(X_i - \bar{X})^2$$

* SSW: Sum of squares for within groups

$$\text{formula} \quad SSW = \sum (X_i - \bar{X})^2$$

5) Compute the degrees of freedom

$$df_B = K - 1$$

↓
no. of groups

$$df_W = N - K$$

↓
no. of observations
 $K =$ " " groups

6) Compute the mean squares

MSB : Mean squares for between groups

$$MSB = \frac{SSB}{df_B}$$

MSW : Mean squares for within groups

$$MSW = \frac{SSW}{df_W}$$

7) Find the F-statistic value

$$F_s = \frac{\text{B/W Group variance}}{\text{within Group variance}} = \frac{MSB}{MSW}$$

8) Compare the F-statistic value with F-critical value

9) If $F_s \geq F_c \rightarrow$ reject H_0

If $F_s < F_c \rightarrow$ fail to reject H_0

Q:- Consider an experiment to study the effect of 3 different levels of a factor on a response (3 levels of a factor on a plant growth). If we add 6 observations for each level we could write the outcome of the experiment in a table like this where a_1, a_2, a_3 are the 3 levels of factor being studied. Test at $\alpha = 0.05$

a_1	a_2	a_3
6	8	13
8	12	9
4	7	11
5	11	8
3	6	7
4	8	12

$$\bar{x}_1 = 5 \quad \bar{x}_2 = 9 \quad \bar{x}_3 = 10$$

$$\bar{x} = \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3}{3} = 8$$

Sol:-

$$H_0: \mu_1 = \mu_2 = \mu_3$$

$$H_1: \mu_1 \neq \mu_2 \neq \mu_3$$

$$\alpha = 0.05$$

$$3) SSB = 2n(x_1^2 + x_2^2 + x_3^2)$$

$$= 6(5^2 + 9^2 + 10^2)$$

$$= 6(25 + 81 + 100)$$

$$= 54 + 486 + 600$$

$$= 1140$$

$$df_B = k - 1 = 3 - 1 = 2$$

$$MSB = \frac{SSB}{df_B}$$

$$SSW = \sum (x_i - \bar{x})^2$$

$$= 6 + 9 + 1$$

$$4 + 9 + 1$$

$$= 20$$

$$SSW = 20$$

$$df_W = 18$$

$$MSW = \frac{SSW}{df_W}$$

$$F_s = \frac{MSB}{MSW}$$

$$F(2, 18)$$

$$df_W$$

15

ANOVA

$H_0: \mu_1 = \mu_2 = \mu_3$ (all means are equal)

$H_1: \mu_1 \neq \mu_2 \neq \mu_3$ (at least one mean is different)

$$\alpha = 0.05$$

$$SSB = \sum n_i (\bar{x}_i - \bar{x})^2$$

$$= 6(8-8)^2 + 6(7-8)^2 + 6(6-8)^2$$

$$= 6(0) + 6(1) + 6(4)$$

$$= 0 + 6 + 24$$

$$= 30$$

$$df_B = K - 1 = 3 - 1 = 2$$

$$MSB = \frac{SSB}{df_B} = \frac{30}{2} = 15$$

$$SSW = \sum (x_i - \bar{x})^2$$

$$1 + 9 + 1 + 0 + 4 + 1 + 1 + 9 + 0 + 4 + 9 + 1 + 9 + 1 + 1 + 4$$

$$4 + 9 + 4 = 68$$

$$= 68$$

$$SSW = 68$$

$$df_W = n - K = 18 - 3 = 15$$

$$MSW = \frac{SSW}{df_W} = \frac{68}{15} = 4.5$$

$$F_3 = \frac{MSB}{MSW} = \frac{15}{4.5} = 3.3$$

$$F(\alpha, df_B, df_W)$$

$$\alpha = 0.05$$

$$df_B = 2$$

df_W

5

3.68

$$F_c = 3.68$$

$$F_3 = 3.3$$

$$F_3 > F_c$$

$$3.3 > 3.68$$

Reject null hypothesis



$$F_0 = \frac{\text{B/w group variance}}{\text{within group variance}}$$

$$\frac{\sum (x_i - \bar{x})^2}{n-k}$$

$$\frac{\sum n_i (\bar{x}_i - \bar{x})^2}{k-1}$$

Q- Using the following data, perform one-way ANOVA

group1	group2	group3
51	23	54
45	43	76
33	21	74
45	43	87
67	45	56

Q- Using one-way ANOVA, perform test at $\alpha = 0.01$ for the given data

sample size (n)	mean	std
30	50.76	10.45
30	45.32	12.76
30	53.67	11.46

Sol-

1) Frame H_0 & H_1

$$H_0: \mu_1 = \mu_2 = \mu_3$$

$$H_1: \mu_1 \neq \mu_2 \neq \mu_3$$

2) choose the significance level

$$\alpha = 0.01$$

3) find the grand mean

$$\bar{x} = \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3}{3} = \frac{50.76 + 45.32 + 53.67}{3} = 49.91$$

find the sum of squares for between groups

$$SSB = \sum n_i (\bar{x}_i - \bar{x})^2$$

$$= 30(50.76 - 49.91)^2 + 30(45.32 - 49.91)^2 + 50(53.67 - 49.91)^2$$

$$= 1949.69 + 1360.59$$

$$df_B = 3 - 1 = 2$$

$$MSB = \frac{SSB}{df_B} = \frac{1360.59}{2} = 680.299 \approx 681$$

$$SSW =$$

$$S^2 = \sum_{i=1}^3 \frac{\sum (x_i - \bar{x})^2}{n_i - 1}$$

$$\sum_{i=1}^3 \frac{\sum (x_i - \bar{x})^2}{n_i - 1} = (n-1) S^2$$

$$= 29(10.45)^2 + 29(12.76)^2 + 49(11.46)^2$$

$$= 14323.83$$

$$df_W = n - k = 110 - 3 = 107$$

$$MSW = \frac{SSW}{df_W} = \frac{14323.83}{107} = 133.867$$

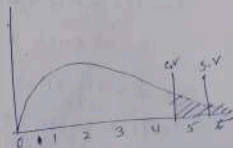
$$F_{\alpha} = \frac{MSB}{MSW} = \frac{680.299}{133.867} = 5.08$$

$$F_{\text{Statistic}} > F_{\text{Critical Value}}$$

$$(5.08) > (4.71)$$

$$F_{(0.05, 2, 107)}$$

Reject H_0



$$F_3 = \frac{\text{B/w group variance}}{\text{within group variance}}$$

$$= \frac{2(\sum (x_i - \bar{x})^2)}{n-k}$$

An education researcher is comparing 4 different algebra curriculum. 80 grade students are randomly assigned to one of the four groups. Their state achievement test scores are compared at the end of the year. Use the appropriate statistical procedure to determine whether the curriculum differ with respect to maths assignments. Test at $\alpha = 0.05$.

sample size (n)	mean	std
50	170.5	14.5
50	188.3	12.8
50	169.6	17.7
50	172.8	16.8

1) Frame H_0 & H_1

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$$

$$H_1: \mu_1 \neq \mu_2 \neq \mu_3 \neq \mu_4$$

$$3) \bar{x} = \frac{x_1 + x_2 + x_3 + x_4}{4} = 170.3$$

Find the sum of squares for b/w groups

$$SSB = \sum n(x_i - \bar{x})^2$$

$$= 50(170.5 - 170.3)^2 + 50(188.3 - 170.3)^2 + 50(169.6 - 170.3)^2 + 50(172.8 - 170.3)^2$$

$$= 2 + 200 + 24.5 + 312.5$$

$$= 539$$

$$dfo = 4 - 1 = 3$$

$$MSB = \frac{SSB}{dfo} = \frac{539}{3}$$

$$SSW = \sum \sum (x_i - \bar{x})^2$$

$$= 49(14.5)^2 + 49(12.8)^2 + 49(17.7)^2 + 49(16.8)^2$$

$$= 47511.3$$

$$dfw = n - k = 80 - 4 = 76$$

$$MSW = \frac{SSW}{dfw}$$

$$F_{\text{statistic}} = \frac{MSB}{MSW}$$

$$F_{(0.05, 3, 76)}$$

failed

$$MSB = \frac{SSB}{dfb} = \frac{539}{3} = 179.66$$

$$SSW \Rightarrow s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

$$\begin{aligned} \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} &= (n-0.5)^2 \\ &= 49(14.5)^2 + 49(12.8)^2 + 49(17.7)^2 + 49(16.6)^2 \\ &= 47511.3 \end{aligned}$$

$$dfw = n - K \Rightarrow 200 - 4 = 196.$$

$$MSW = \frac{SSW}{dfw} = \frac{47511.3}{196} = 242.4.$$

$$F_{statistic} = \frac{MSB}{MSW} = \frac{179.6}{242.4} = 0.74.$$

$$F(0.05, 3, 196) = 2.6$$

failed to reject null hypothesis.

F-statistic $\leq$ F-critical
0.74 $\leq$ 2.6

